

David Park

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EDUCATION

University of Illinois at Urbana-Champaign

Expected Graduation: May, 2027

Master of Science in Industrial Engineering

Bachelor of Science in Industrial Engineering, Minor in Computer Science

RELATED COURSEWORK

Analysis of Data, Deterministic Models in Optimization, Production Planning & Control, Stochastic Processes & Applications, Deep Learning, Business side of Engineering, Discrete Structures, Data Structure, Financial Engineering

WORK EXPERIENCE

Samsung Electronics

June 2024 – August 2024

DRAM Application Engineer Intern

- Adjusted memory eye values by tuning ODT registers to stabilize signal margins between Samsung DDR5 SODIMM and Intel/AMD boards, optimizing system performance and improving compatibility by 10%
- Investigated the impact of ODT register values on memory eye margins in new models to identify optimal settings that enhanced next-generation product performance utilizing Python and Bayesian method

Industrial & Mathematical Data Research Center, Seoul National University

June 2023 – August 2023

Topological Data Analytics Researcher

- Collected and preprocessed over 50,000 tabular data sets on students' academic records and socio-economic backgrounds to prepare for large-scale structural analysis
- Modeled tabular data as a high-dimensional point cloud using Topological Data Analysis (TDA) to uncover noise-robust global structures and relationships between background factors and academic performance
- Combined TDA with density-based clustering to extract interpretable topological features, revealing recurring socio-economic patterns linked to distinct performance trends

PROJECTS

Real-Time Portfolio VaR Monitoring System

January 2026 – May 2026

- Developed a near real-time portfolio Value-at-Risk (VaR) monitoring system for a five-asset portfolio updating 95% and 99% VaR estimates at 1-minute intervals using intraday market data replay
- Benchmarked rolling covariance, EWMA, and Monte Carlo GBM VaR models, quantifying latency-accuracy trade-offs and achieving sub-millisecond parametric VaR computation
- Backtested VaR performance using the Kupiec test across 498 observations, validating model calibration with 7 breaches and visualizing exceedances in a Plotly/Dash dashboard

Electricity Price Forecasting & Model Evaluation

September 2025 – December 2025

- Architected an end-to-end forecasting framework on 25,000+ hourly observations, benchmarking Ridge Regression, FFNN, LSTM, and GRU utilizing expanding-window validation
- Formulated 20+ temporal and exogenous features while leading 50+ controlled experiments with systematic hyperparameter tuning to improve out-of-sample performance
- Translated prediction outputs into operational decision-making by improving demand planning accuracy, reducing RMSE by 18% through rigorous error analysis

UPS Staff Demand Analysis & Scheduling Tool Design

August 2024 – December 2024

- Built a mathematical staffing optimization model for UPS's Ocean Products management by forecasting monthly labor demand under uncertainty
- Performed demand forecasting and time studies from UPS operational videos, translating process insights into quantitative parameters for the model
- Optimized labor allocation and quantified the economic benefits of optimization, achieving over 20% improvement in work performance and more than 15% cost reduction

NLP Models for Cyberbullying Detection

September 2023 – March 2024

- Conducted research on detecting cyberbullying in social media comments using NLP models like BERT, RoBERTa, and XLNet by classifying aggressive language from labeled datasets by leading a team of three
- Applied under-sampling techniques to address dataset imbalances and trained the models to distinguish between harmful and non-harmful comments based on linguistic patterns
- Selected as a presenter at the MRC conference and presented findings, highlighting RoBERTa as the most effective model

SKILLS

- Programming:** Python (data analysis, ML, time-series forecasting), SQL, C++
- Data & AI:** feature engineering, data preprocessing, anomaly detection, statistical modeling
- Domain:** manufacturing systems, demand forecasting, optimization under uncertainty